

Vector Stores

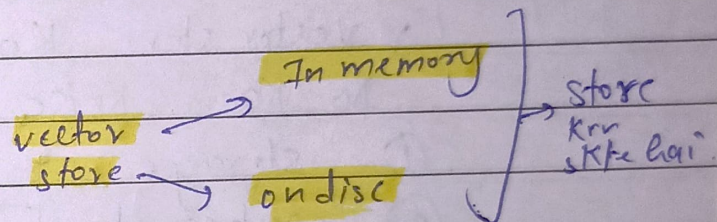
is a system designed to store and retrieve data represented as numerical vectors.

- vector store = ek special database Jo text ko numbers (embeddings) mein convert karke store karta hai, taaki semantic search ho ske !!

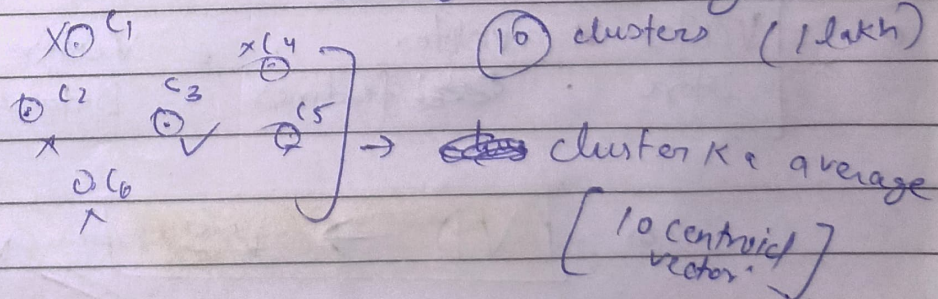
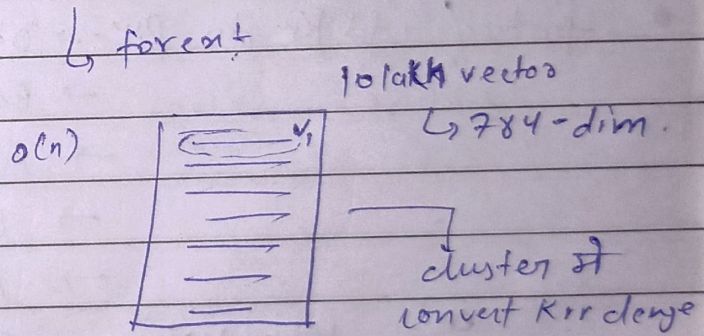
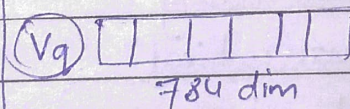
- Normal database Keyword match karta hai
- Vector space meaning match karta hai

Key features :-

- 1) Storage Memory
- 2) Similarity Search
- 3) Indexing → fast similarity search
- 4) CRUD operation



adding new vectors,
retrieving them,
updating,
removing/deleting.



Use-cases ↓

- 1) Semantic Search
- 2) Recommendation system
- 3) RAG
- 4) Image/multimedia Search.

#

Vector store V/S Vector Database• big Difference

↳ Vector store Ko system mein store hai Jo do kam krke de skta hai

- ① storage
 - ② retrieval
- Isko vector store

• features add kr do like → Distributed

database

- Backup and restore
- ACID Transactⁿ.
- concurrency
- Authenticaⁿ

→ Ye add kr diya
tho
vector
database.

ex: ~~Milvus~~ Milvus, Qdrant, Weaviate.

$$\left[\begin{array}{l} \text{Vector + vector} \\ \text{store} \cdot \text{Data features} \end{array} = \text{Vector Database} \right]$$

$$\text{VD} \rightarrow \text{VS} \quad \checkmark$$

$$\text{VS} \rightarrow \text{VD} \rightarrow \text{X}$$

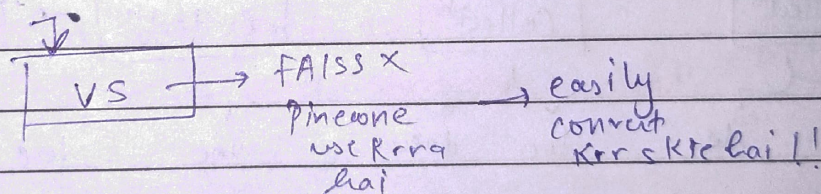
◦ Vector stores in langchain :-

- Supported Stores :- langchain Integrate with multiple vector stores
- Common Interface
- Metadata Handling +

↓
(FAISS → facebook ka
Pinecone
Chroma
Qdrant
Weaviate) etc .

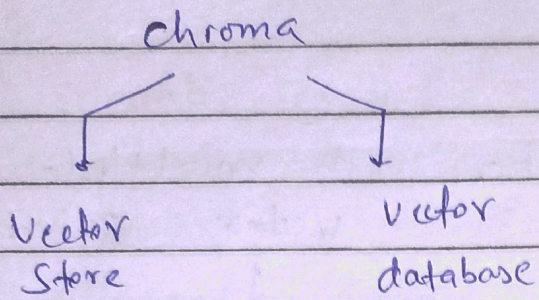
from - documents(...) or from_texts(----)
add - documents(...) or add_texts(----)
similarity_search(query, k=----)

Metadata - based filtering .

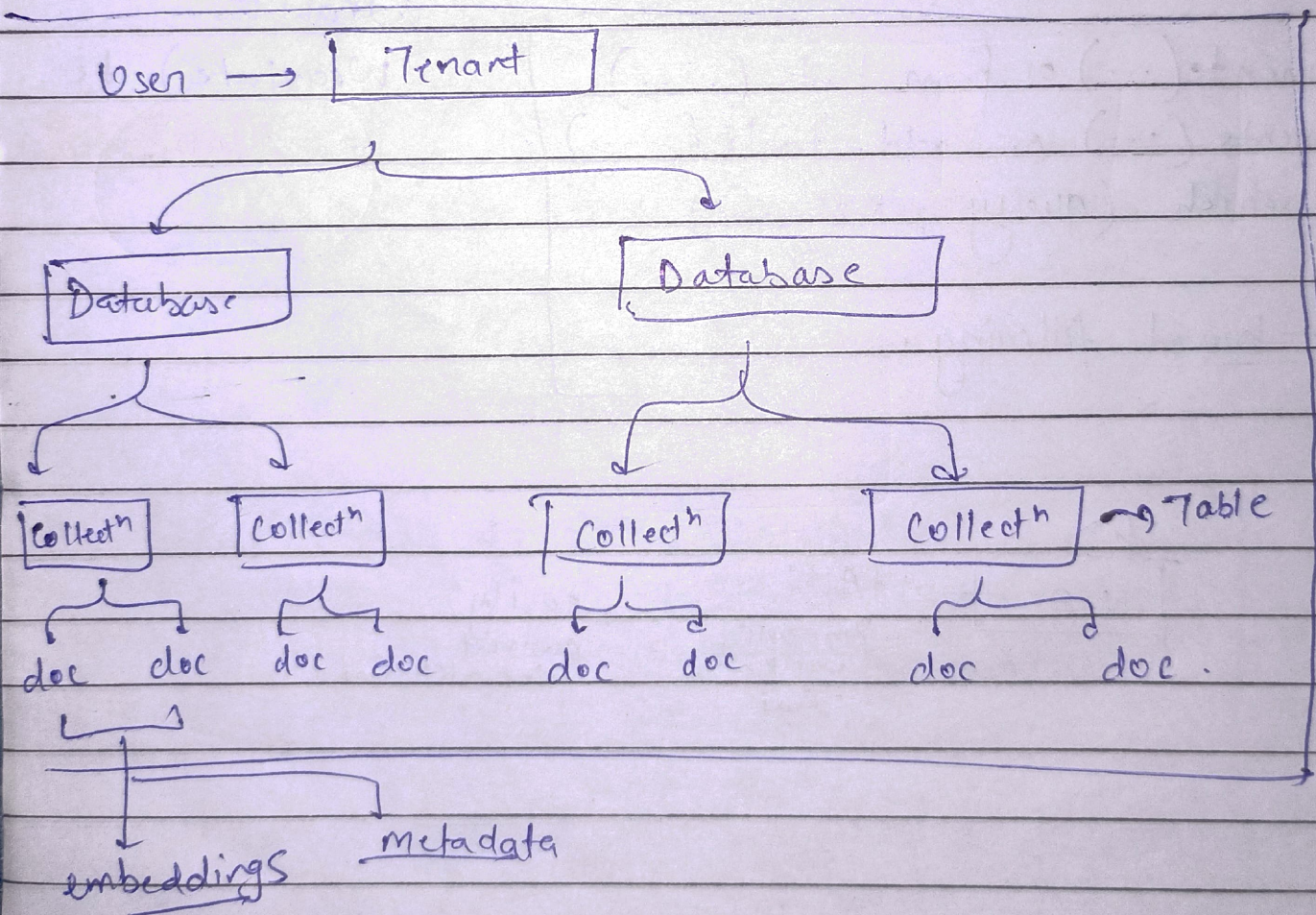


chroma Vector store :-

- chroma is a light weight open source vector database that is especially friendly for local development and small to medium scale production needs .



chroma Tenancy and DB Hierarchy



Code :-

```
vector_store = chroma (
    embedding_function = OpenAI embeddings(),
    persist_directory = 'chroma_db',
    collection_name = 'sample'
```

3:

① Add document to vector Database

```
vector_store.add_documents (docs)
```

→ Tent data ko embeddings ke through database me save karta hai

② Similarity Search Query :-

```
vector_store.similarity_search (
    query = "who among these are a bowler",
    k = 2
```

)

→ Top K most relevant documents return karta hai

③ Similarity Search with Score

```
vector_store.similarity_search_with_score (
```

```
    query = "who among these are a bowler",
```

```
    k = 2
```

)

• Konsa result kitna relevant hai wo numeric score me milta hai

④ Metadata filtering Query

```
vector_store.similarity_search_with_score (
```

```
    query = " ",
```

```
    filter = { "team": "CSK" }
```

)

• Sirf filter match karne wala document return karta hai